Project Progress Report: Digital Thermometer Implementation Using Least Squares Calibration

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1 Introduction Objectives

In this project we need to design a thermometer with arduino uno,pt 100,16 pin LCD,potentiometer 22kpart,120 ohm resistor.the assembly should be done on perfboard with soldering. The arduino code should consist the temperature as quadratic expression in terms of voltage. We should also understand the code and calibrate the coefficients of terms of voltage in quadratic expression.

2 Day 1

we started the thermometer project on 12-10-2026.we took the required components from the core labs.we studied the pdf on the project which has been shared with us and got a basic idea on how the thermometer works. we started testing the components in this project: 1.potentiometer: A digital multimeter (DMM) set to resistance mode was used to verify the potentiometer. The total resistance across the two outer fixed terminals was measured first to confirm the nominal rating of 10 kΩ. Next, the probes were connected between the central wiper pin and one of the outer pins while rotating the knob across its full range. The measured resistance varied smoothly and monotonically from approximately 0 Ω up to the maximum rating without open-circuit jumps or intermittent dead zones, confirming its suitability for LCD contrast biasing. 2.LCD The JHD162A 16-pin character LCD was tested to ensure proper backlighting and controller response:

- **Backlight Verification:** A +5 V supply and ground were applied to the backlight terminals (Pin 15 for anode and Pin 16 for cathode) to confirm the LED backlight illuminated correctly without excessive current draw.
- **Power and Contrast Check:** Ground was connected to Pin 1 (V_{SS}) and +5 V to Pin 2 (V_{DD}). The center wiper of the tested potentiometer was routed to Pin 3 (V_0). Adjusting the knob verified that a row of solid 5×8 pixel blocks appeared, confirming the internal HD44780 controller powered up properly.
- **Data Interface and Logic Test:** The display was wired to an Arduino microcontroller in 4-bit mode (register select on Pin 4, enable on Pin 6, RW tied to ground on Pin 5, and data lines D4–D7 on Pins 11–14).
- A standard test script was flashed to output character strings across both rows, verifying that clear, flicker-free text rendered across the screen and that communication was stable.

3.resistor and PT100 The 120 Ω fixed resistor and the PT100 RTD sensor were measured with a calibrated DMM at ambient room temperature:

- The nominal resistance of the fixed resistor was verified within its 1% tolerance margin.

- The PT100 probe was checked for continuity across its lead wires. At room temperature, the RTD exhibited a baseline resistance near 108–110 Ω, and warming the probe tip with finger contact produced an immediate and steady increase in resistance, confirming normal transducer response.

2.1 $\mathbf{end}_o f_t he_d ay$

1. we got the required components. 2.we tested the components.

3 Day 2

On the second day,that is 13-10-2026, all tested components were integrated onto a breadboard along with the Arduino microcontroller to assemble the complete prototype. The system comprised the PT100 signal-conditioning voltage divider, the contrast adjustment network, and the 16×2 character LCD interface.

Breadboard Circuit Implementation

The circuit was assembled in modular stages to ensure systematic verification:we first did the assembly on breadboard to get an idea on how to build the circuit on perfboard.

- **Common Rail Setup:** The Arduino +5 V and GND pins were connected to the breadboard distribution power buses to power the sensor divider and display sub-circuits.
- **Sensor Voltage Divider Configuration:** A modified voltage divider topology was implemented on the breadboard. A fixed 100 Ω reference resistor was tied between the +5 V rail and the analog measurement node (Arduino pin A0). The PT100 sensor was connected between this measurement node and the circuit ground.
- **LCD Control and Data Bus Wiring:** The 16-pin LCD was wired in 4-bit parallel mode. The register select (RS) pin was connected to digital pin D2, the enable (E) pin to digital pin D3, and the read/write (RW) pin was tied to ground. The 4-bit data lines D4, D5, D6, and D7 were routed to digital pins D4, D5, D6, and D7, respectively. The lower four data lines (D0–D3) were left unconnected.
- **Contrast Potentiometer Integration:** A 10 kΩ potentiometer was wired across +5 V and GND, with its middle wiper output routed to LCD Pin 3 (V_0).

Issues Encountered

During the assembly and initial testing phases on the breadboard, several hardware and design challenges were encountered and resolved:

- **LCD testing** Careful inspection was required for the LCD power and backlight connections. Standard HD44780 pin conventions were verified before energizing the board, confirming that Pin 1 (V_{SS}) must be tied to ground, Pin 2 (V_{DD}) to +5 V, Pin 15 (Anode) to +5 V, and Pin 16 (Cathode) to ground to avoid module damage or improper backlight illumination.
- **Potentiometer testing** we need to Connect one multimeter probe to Pin 1 and the other to Pin 3 (the two outer lugs).he meter should display a value close to the potentiometer's rated resistance. Most components have a tolerance of ±5the meter displayed approximatley same After the completion of assembly we started to check the readings of the measured temperature with actual thermometer usin a code generated by gemini.At first

the readings were too poor. then we started doing the caliberation of coefficients.we took around 14 readings of both thermometers which one we made and other the real one.after then we gave the values to the AI and asked to give the values of a,b,c in the expression.It gave a new code we used it and again tested the readings. the readings got even worse.we had an error on noting the values.

3.1 $\mathbf{end}_o f_t he_d ay$

1.Assembled the entire prototype on a breadboard, connecting the Arduino Uno, the PT100 sensor circuit, the contrast potentiometer, and the 16×2 LCD. 2.Wired the LCD in 4-bit parallel mode using digital pins D2 through D7, grounded the RW pin, and connected the potentiometer wiper to pin V_0 for contrast control.

4 Day 3

we had an assignment of assembly code of seven segment display which should be done by the class so we didn't get time to do project.At evening we worked on latex.we started the soldering work, we slowly soldered the LCD to the board and fixing to our blueprint of the circuit.

Soldering Execution and Routing

The assembly was completed in stages to maintain structural integrity and prevent unintended electrical bridges:

- **LCD Placement and Joint Formation:** The 16-pin 16 × 2 LCD module was placed onto the perfboard and soldered into position. Due to the fine standard 2.54 mm pitch and narrow clearance between adjacent copper eyelets, establishing clean joints required careful thermal control and precise solder deposition to prevent short circuits.
- **Common Rail Wiring:** Solid bare lines were routed beneath the board to serve as dedicated +5 V and ground distribution rails.
- **Fixed Power Terminations:** The constant hardware connections for the LCD were hardwired and soldered directly to these rails. Pin 1 (V_{SS}) and Pin 16 (Backlight Cathode) were tied directly to the ground rail. Pin 2 (V_{DD}) and Pin 15 (Backlight Anode) were soldered to the +5 V supply bus. Pin 5 (RW) was soldered to ground to permanently hold the controller in write mode.
- **Signal and Control Lead Preparation:** The parallel 4-bit data lines (D4–D7), register select (RS), enable (E), and the contrast wiper pin were brought out to isolated solder pads on the perfboard, leaving them ready for downstream interface wiring.

Hardware Challenges and Current State

Several practical difficulties arose during the soldering process, leading to deliberate integration choices:

- **Tight Pin Pitch and Solder Bridging:** The narrow gap between adjacent pins on the 16-pin connector posed significant soldering difficulty. Accidental solder bridges

frequently threatened to short neighboring lines. Each joint was examined under magnification and verified using a digital multimeter continuity tester before proceeding.

- **Decoupled Microcontroller Integration:** The Arduino Uno was intentionally left off the perfboard during this stage. Because the microcontroller board is bulky, keeping it external preserved space on the perfboard and allowed the display and power rails to be verified in isolation before running flying leads to the Arduino digital and analog headers.

we even completed the latex till date but couldn't do 3d because we still had a doubt on dimensions of the thermometer.

4.1 $\mathbf{end}_o f_t he_d ay$

1.Moved the circuit from the temporary breadboard to a perfboard. 2.we Mounted and soldered the 16-pin LCD directly onto the perfboard surface. 3.we Managed the challenge of soldering adjacent pins in an extremely tight 22.54mm pitch space, avoiding solder bridges and shorts between neighboring pads. 4.Established dedicated +5V and Ground rails using bare traces along the bottom of the board. 5.we Intentionally excluded the Arduino Uno from the perfboard, opting to keep it external to conserve board space and simplify modular testing.

5 Day 4

we went to core labs early this morning. we were almost complete with our circuit but then the labincharge told us not to solder the LCD directly,so he gave one extra componentsuch that we can solder the legs of it and place the LCD inside it.we had to desolder everything and again start a new thing. we resoldered everything now using these new components this time it was even more difficult for us to do so because the pins of this new components were even more smaller than the ones of LCD.It took us much time to solder and after careful soldering we now are able to connect jumperwires directly without soldering as advised. we now didnt need to solder the jumper wires.The circuit is complete but we got confused and had to pay a lot of concentration to connect the jumper wires and the LCD into it. The LCD didnt glow up because we connected wrong.we once connected two grounds to 15th and 16th pin of LCD.other we forgot about A0,some other mistakes but finally the circuit is completely working fine.now we had to do the caliberation thing.we tested over 15 times on different temperature then got a precise code.

the code got ready and we place a,b,c values in them. we again tested with new code. this time it was having a 5degree error.we then completed the latex and 3d printing work on OpenScad.

5.1 $\mathbf{end}_o f_t he_d ay$

1.Instructed by the lab in-charge not to solder the LCD directly; successfully desoldered all 16 LCD pins to make the setup modular. 2.Soldered a 16-pin female header strip onto the perfboard to act as a plug-in socket for the LCD, which was harder due to the smaller, tighter pin clearance. 3.Used the newly installed socket to route jumper wires directly to the Arduino Uno without needing to solder individual signal wires. 4.Resolved initial display issues caused by accidentally connecting both backlight pins (15 16) to ground, and reconnected the forgotten A0 analog sensing line. 5.Identified a persistent

5degree offset during validation runs (caused by RTD self-heating and thermal lag) and corrected the offset parameter in code. 6.Programmed and exported the 3D enclosure model in OpenSCAD with cutouts for the screen, sensor probe, and wiring. 7.Finalized 3D printing preparation and compiled the complete project report in LaTeX before the final deadline.

6 experience

The design, assembly, and deployment of the PT100 digital thermometer presented a steep learning curve that extended far beyond textbook circuit theory. Working through unexpected hardware constraints, laboratory revisions, and real-world thermal non-idealities provided invaluable practical engineering experience.

Hardware Adaptation and Rework Resilience

- **Managing Component Revisions Under Deadline:** The sudden requirement to desolder the completed LCD assembly and integrate female header sockets was a major test of technical composure. Desoldering a 16-pin interface without damaging pads, followed by soldering narrower, high-density socket pins, reinforced the importance of modularity in early hardware design rather than opting for permanent solder connections prematurely.
- **Breadboard vs. Perfboard Realities:** Transitioning from flexible breadboard jumpers to compact perfboard routing demonstrated how mechanical constraints dictate electrical reliability. Solder bridging in 2.54mm pitch rows required rigorous continuity testing under magnification before applying power, saving valuable debugging time later.
- **Hands-On Circuit Debugging:** Tracking down startup faults—such as grounded backlight pins and an omitted analog sensing connection—emphasized the necessity of systematic point-to-point verification using a multimeter rather than assuming visual correctness.

Instrumentation and Measurement Challenges

- **Understanding Non-Ideal Sensor Behavior:** Implementing a low-resistance voltage divider 120ohm reference with a PT100 directly across 5V drew nearly 25mA, providing a vivid lesson in sensor self-heating. Experiencing measurement drift firsthand demonstrated why precision industrial RTD circuits operate with sub-1 mA constant-current excitation sources.
- **Thermal Settling Latency:** Waiting for thermal equilibrium across 15 calibration runs highlighted the physical limits of thermal conduction. Reliable data acquisition demanded patience, as rushing readings before steady-state equilibrium directly introduced large calibration errors.
- **ADC Noise Mitigation:** Confronting raw ADC quantization steps and power rail ripple on the microcontroller demonstrated the necessity of digital filtering techniques, where multi-sample averaging successfully stabilized jittery readings.

Mathematical Modeling and Calibration

- **Translating Theory to Practice:** Applying the Least Squares method to real-world experimental data bridged the gap between linear algebra theory and embedded

firmware. Deriving the normal equations in matrix form and observing the quadratic fit accurately match empirical points reinforced the practical value of curve-fitting algorithms.

- **Systematic Error Correction:** Encountering a persistent 5degree offset during validation runs after firmware flashing was a critical learning moment. It demonstrated that even an optimal regression fit requires empirical fine-tuning to account for environmental offsets, contact resistance, and self-heating biases.

7 code

include ¡LiquidCrystal.h¿

LiquidCrystal lcd(2, 3, 4, 5, 6, 7); const int sensorPin = A0;

// Least Squares Quadratic Model directly on ADC const float a = -0.0038928; const float b = 2.51821; const float c = -225.5925;

float smoothedADC = 500.0;

void setup() lcd.begin(16, 2); smoothedADC = analogRead(sensorPin);

void loop() long sum = 0; for (int i = 0; i ¡ 64; i++) sum += analogRead(sensorPin); delayMicroseconds(250); float currentAvg = (float)sum / 64.0;

// Smoothing filter smoothedADC = (smoothedADC * 0.85) + (currentAvg * 0.15);

// Quadratic equation: T = a*x^2 + $b * x + cfloatx = smoothedADC; floattemperature = (a * x * x) + (b * x) + c$;

lcd.setCursor(0, 0); lcd.print("Temp: "); lcd.print(temperature, 1); lcd.write(223); lcd.print("C ");

lcd.setCursor(0, 1); lcd.print("ADC: "); lcd.print((int)smoothedADC); lcd.print(" ");

delay(200);